ZHIQI HUANG

SHENZHEN, CHINA / FUKUOKA, JAPAN

EMAIL HOMEPAGE SCHOLAR GITHUB

RESEARCH PROFILE

- Research Assistant at CUHK-Shenzhen and M.Phil. student at Waseda University, studying how structured 3D models can turn real observations into useful environments for robot learning.
- My work has progressed from controllable 3D avatars and geometry-aligned relightable materials to predictive modeling of volumetric change. I am now extending these ideas toward task-aware digital cousins and action-conditioned 3D environment models.

RESEARCH INTERESTS

- Task-aware real-to-sim and editable 3D digital cousins for robot learning.
- Action-conditioned 3D transition and world models for robotic manipulation.
- Sim-and-real co-training, reality-gap diagnosis, and performance-based transfer evaluation.

RESEARCH EXPERIENCE

The Chinese University of Hong Kong, Shenzhen — Jul. 2026 - Present

Research Assistant — Shenzhen, China

- Research focus: task-grounded 3D environment models for real-to-sim-to-real robot learning.

EDUCATION

Waseda University — Apr. 2025 - Mar. 2027 (expected)

M.Phil. in Information Architecture — Fukuoka, Japan

- English-taught program; GPA: 3.8 / 4.0

Sun Yat-sen University — 2017 - 2021

B.E. in Software Engineering — Guangzhou, China

- GPA: 3.7 / 4.0

PUBLICATION

SIE3D: Single-Image Expressive 3D Avatar Generation via Semantic Embedding and Perceptual Expression Loss — 2026

IEEE ICASSP 2026; First Author, Corresponding Author

- Authors: Zhiqi Huang, Dulongkai Cui, Jinglu Hu
- Generates an editable 3D Gaussian head avatar from one image while preserving identity and enabling language-level control over expressions and accessories.
- Links: Project, IEEE Xplore, arXiv, Code

MANUSCRIPTS UNDER REVIEW

Controllable PBR material generation from long-form descriptions 2026

First Author; AAAI 2027 (Under Review)

- Generates relightable PBR channels by retaining detailed material specifications and routing them with surface geometry.
- Extends controllable 3D modeling to geometry-aligned, physically based object appearance, with relevance to the visual-observation side of real-to-sim.

Deform-then-edit forecasting for longitudinal 3D CT 2026

First Author; AAAI 2027 (Under Review)

- Forecasts localized volumetric change while explicitly preserving stable surrounding anatomy.
- Introduces a history-conditioned deform-then-edit transition model; it provides a methodological basis for future action-conditioned 3D transitions but does not model robot actions itself.

INDUSTRY EXPERIENCE

4399 Games 2023 - 2024

Senior Graphics Engineer Guangzhou, China

- Led a 3-5 person rendering team and owned the rendering roadmap for *Era of Conquest* on mobile and PC.
- Developed reusable asset, material, and rendering pipelines for mobile, PC, and web projects.

4399 Games 2021 - 2023

Graphics Engineer Guangzhou, China

- Built and optimized the cross-platform real-time rendering pipeline for *Era of Conquest*, including shaders, PBR materials, and mobile performance profiling.

TECHNICAL SKILLS

- Graphics and systems: C++, C#, GLSL/HLSL; Unity, Vulkan, OpenGL, real-time rendering, PBR, cross-platform optimization.
- Machine learning and 3D: Python, PyTorch, 3D Gaussian Splatting, diffusion models, CLIP/LongCLIP, mesh and material processing, multi-view evaluation, volumetric modeling.

LANGUAGES AND HONORS

- Chinese: Native (Mandarin and Cantonese)
- English: Professional working proficiency (TOEFL iBT: 90)
- Outstanding Student Scholarship (Third Prize), Sun Yat-sen University, 2018-2019